

# Liver Lesion Timeline Report

Three CT dates · lesion trends · hemodynamic normalization · vessel proximity

25 DEC 2025

**9.78%**

166.82 mL segmented tumor

19 JAN 2026

**10.22%**

170.34 mL segmented tumor

23 AUG 2026

**3.17%**

46.56 mL segmented tumor

JAN → AUG TUMOR VOLUME

**-72.7%**

170.34 → 46.56 mL

LATEST TRACKED GROUPS

**9 / 16**

12 separated residual foci

**Important:** This report is generated from automated masks and image registration. It is not a radiologist-signed diagnosis, formal RECIST assessment, tumor-viability test, or surgical-planning document.

# Substantially lower automated burden on the latest CT

TUMOR VOLUME

-72.7%

January → August

TUMOR BURDEN

3.17%

10.22% → 3.17%

REFERENCE SIMILARITY

91.6%

High Jan–Aug comparability

RESIDUAL GROUPS

9

11 separated foci

## Bottom-line image-only observations

- 01

The image-only automated models show substantially less segmented liver tumor volume in August than January, driven mainly by L01 and L02. This is compatible with interval treatment response, but protocol differences, registration, and segmentation uncertainty require radiologist confirmation.
- 02

The dominant segment 3 group decreased from 106.21 mL to 17.15 mL in the automated masks.
- 03

The segment 8 group is fragmented into two residual foci totaling 25.08 mL and lies close to automated hepatic-vessel and IVC masks.
- 04

Both scans were acquired 90 seconds after an identical 80 mL iodine injection at 1.7 mL/s. VNC-corrected internal liver and blood-pool references are also similar. The similarity score is an internal technical quality indicator, not a clinical probability.

## Timeline

- 25 Dec 2025

Tumor 166.82 mL · burden 9.78%
- 19 Jan 2026

Tumor 170.34 mL · burden 10.22%
- 23 Aug 2026

Tumor 46.56 mL · burden 3.17%

05

CT attenuation and low-attenuation fractions are imaging proxies only. VNC correction and internal-reference normalization reduce contrast delivery and hemodynamic bias, but portal-venous CT still cannot determine whether tumor cells are alive or establish true necrosis.

Interpretation boundary: The direction and magnitude of change are model-derived. Different scanners, slice thickness, contrast timing, reconstruction, registration, and contour behavior can change volume and attenuation. Review the original DICOM studies with the treating radiologist before clinical decisions.

# L01 · reference liver segment 3

MARKED DECREASE

LONG AXIS

**64.5 → 33.0 mm**

Change -48.8%

SEGMENTED VOLUME

**106.21 → 17.15 mL**

Change -83.8%

VNC-CORRECTED ENHANCEMENT

**26.7 → 6.2 HU**

Vs local liver: 47.6% → 12.6%

## Automated assessment

Marked automated volume decrease since January.

**Minimal enhancement proxy (<10 HU):** 13.4% → 61.7%

**Absolute <40 HU:** 5.2% → 33.3%

**Hepatic Vessels:** 2.6 mm

**Portal Vein:** 4.9 mm

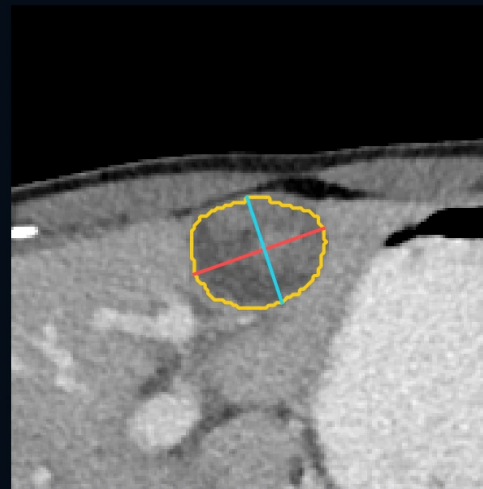
**Confidence:** high

## L01 · automated comparison

19 Jan 2026  
64.5 × 51.1 mm · 106.21 mL



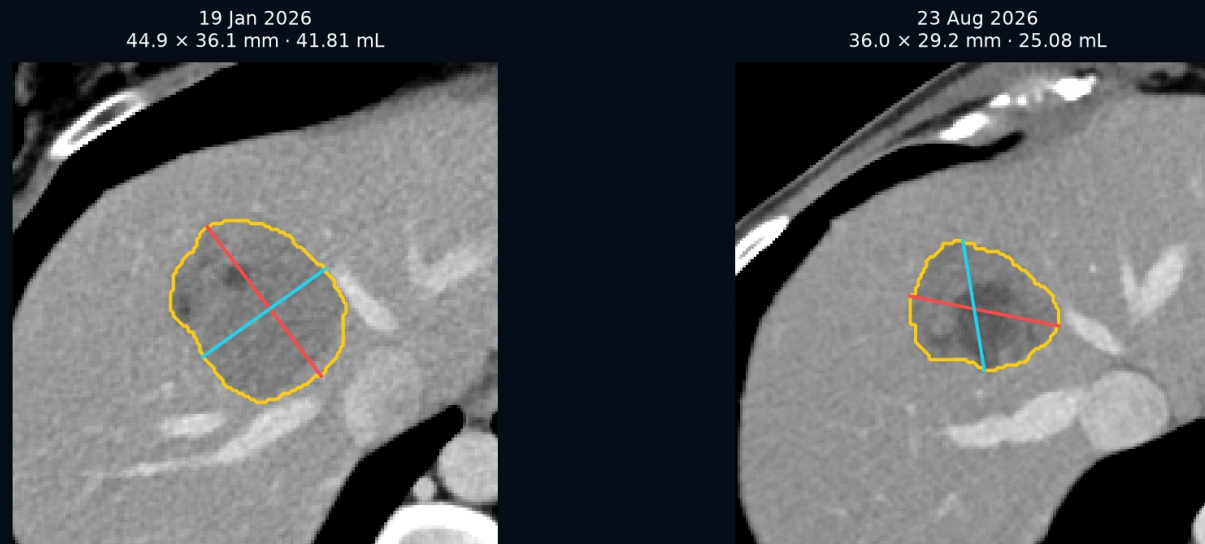
23 Aug 2026  
32.7 × 26.4 mm · 17.15 mL



Yellow: automated boundary · Red: long axis · Cyan: perpendicular axis · Verify with a radiologist

# L02 · reference liver segment 8

## L02 · automated comparison



Yellow: automated boundary · Red: long axis · Cyan: perpendicular axis · Verify with a radiologist

MARKED DECREASE

LONG AXIS

**45.0 → 36.0 mm**

Change -20.0%

SEGMENTED VOLUME

**41.81 → 25.08 mL**

Change -40.0%

VNC-CORRECTED ENHANCEMENT

**47.5 → 31.3 HU**

Vs local liver: 81.4% → 63.0%

### Automated assessment

Automated volume decrease since January. The residual mask is split into 2 foci.

**Minimal enhancement proxy (<10 HU):** 0.3% → 10.3%

**Absolute <40 HU:** 0.4% → 10.1%

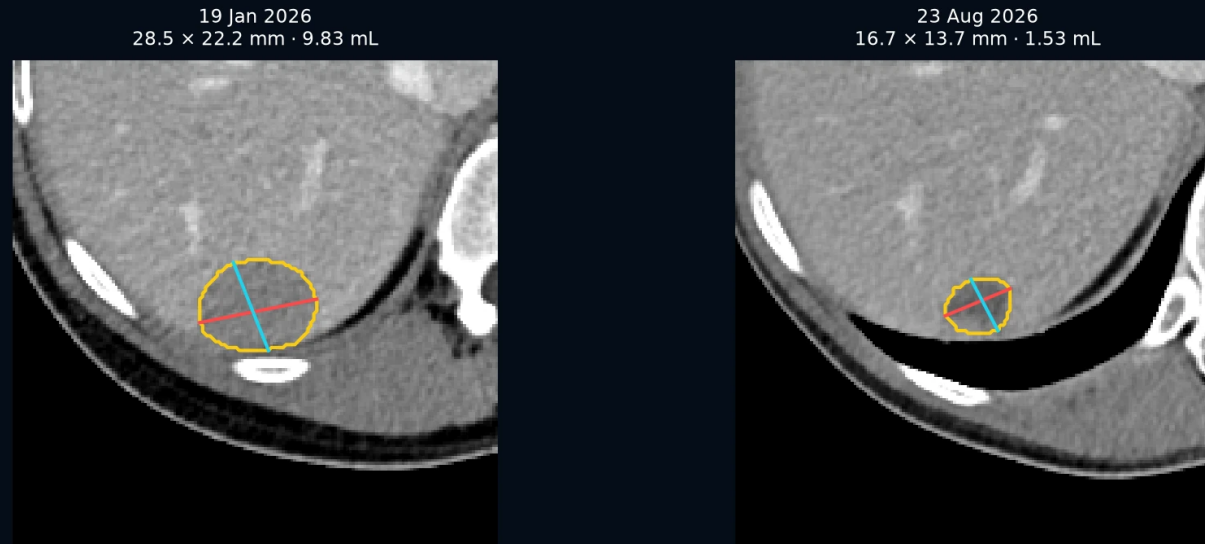
**Hepatic Vessels:** 0.6 mm

**Ivc:** 2.3 mm

**Confidence:** high

# L03 · reference liver segment 7

## L03 · automated comparison



Yellow: automated boundary · Red: long axis · Cyan: perpendicular axis · Verify with a radiologist

MARKED DECREASE

LONG AXIS

**28.5 → 16.7 mm**

Change -41.4%

SEGMENTED VOLUME

**9.83 → 1.53 mL**

Change -84.4%

VNC-CORRECTED ENHANCEMENT

**41.1 → 12.0 HU**

Vs local liver: 71.9% → 23.5%

### Automated assessment

Marked automated volume decrease since January.

**Minimal enhancement proxy (<10 HU):** 0.6% → 44.8%

**Absolute <40 HU:** 0.0% → 18.9%

**Hepatic Vessels:** 14.4 mm

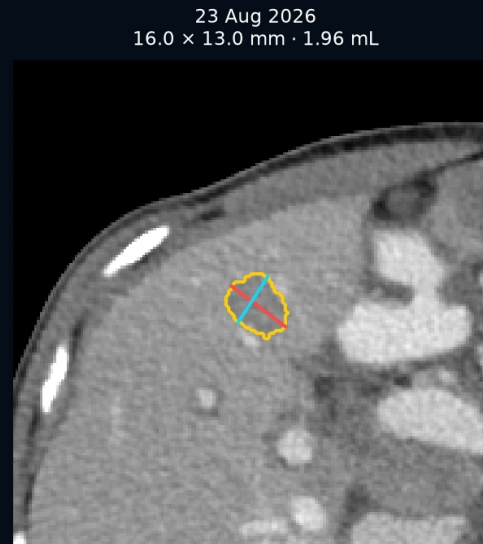
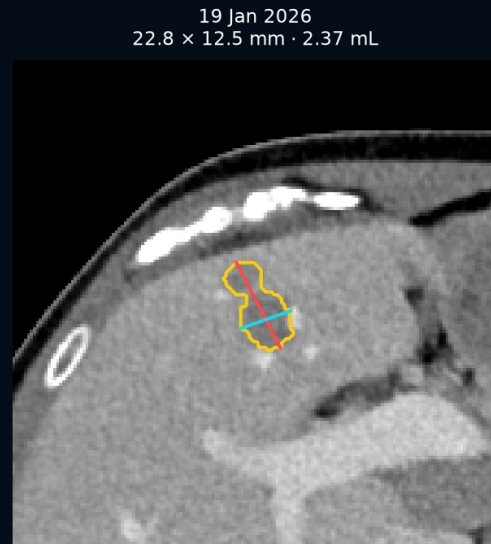
**Portal Vein:** 51.3 mm

**Confidence:** moderate

# L04 · reference liver segment 4

REVIEW CHANGE

## L04 · automated comparison



Yellow: automated boundary · Red: long axis · Cyan: perpendicular axis · Verify with a radiologist

LONG AXIS

**22.9 → 16.0 mm**

Change -30.1%

SEGMENTED VOLUME

**2.37 → 1.96 mL**

Change -17.6%

VNC-CORRECTED ENHANCEMENT

**33.8 → 45.9 HU**

Vs local liver: 63.9% → 87.0%

### Automated assessment

No large automated volume change. The residual mask is split into 2 foci.

Minimal enhancement proxy (<10 HU): 3.6%  
→ 0.1%

Absolute <40 HU: 0.3% → 0.0%

Hepatic Vessels: 1.4 mm

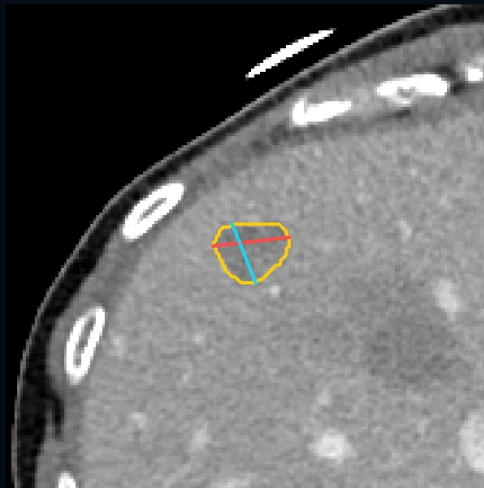
Portal Vein: 17.8 mm

Confidence: moderate

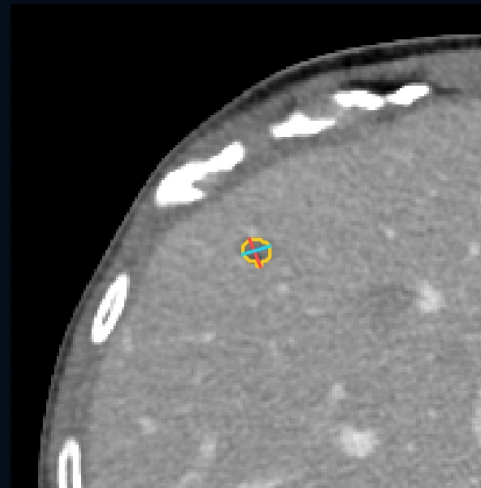
# L05 · reference liver segment 8

## L05 · automated comparison

19 Jan 2026  
18.3 × 14.5 mm · 1.89 mL



23 Aug 2026  
6.9 × 6.7 mm · 0.12 mL



Yellow: automated boundary · Red: long axis · Cyan: perpendicular axis · Verify with a radiologist

MARKED DECREASE

LONG AXIS

**18.3 → 6.9 mm**

Change -62.3%

SEGMENTED VOLUME

**1.89 → 0.12 mL**

Change -93.8%

VNC-CORRECTED ENHANCEMENT

**43.8 → 29.7 HU**

Vs local liver: 80.6% → 60.4%

### Automated assessment

Marked automated volume decrease since January.

Minimal enhancement proxy (<10 HU): 0.1% → 11.5%

Absolute <40 HU: 0.0% → 5.7%

Hepatic Vessels: 4.5 mm

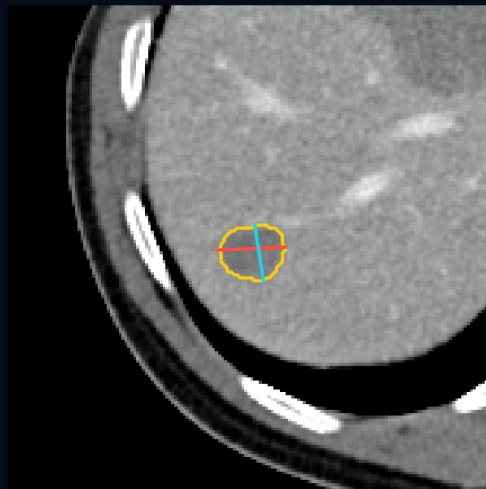
Portal Vein: 31.3 mm

Confidence: moderate

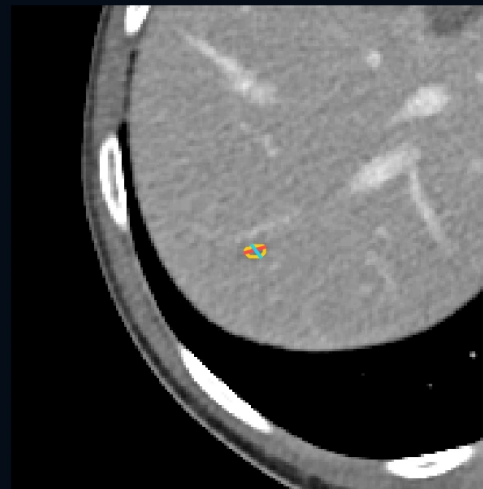
# L06 · reference liver segment 7

## L06 · automated comparison

19 Jan 2026  
16.1 × 13.4 mm · 1.86 mL



23 Aug 2026  
4.7 × 3.0 mm · 0.04 mL



Yellow: automated boundary · Red: long axis · Cyan: perpendicular axis · Verify with a radiologist

MARKED DECREASE

LONG AXIS

**16.1 → 4.7 mm**

Change -70.8%

SEGMENTED VOLUME

**1.85 → 0.04 mL**

Change -98.0%

VNC-CORRECTED ENHANCEMENT

**34.8 → 32.3 HU**

Vs local liver: 65.8% → 72.6%

### Automated assessment

Marked automated volume decrease since January.

Minimal enhancement proxy (<10 HU): 0.4% → 4.8%

Absolute <40 HU: 0.0% → 0.0%

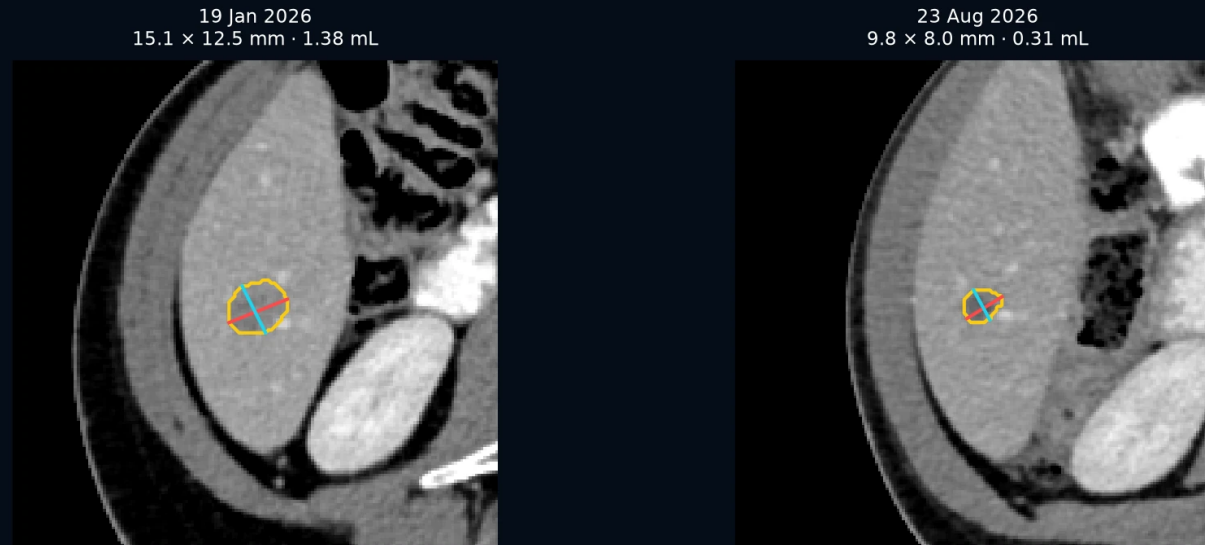
Hepatic Vessels: 13.9 mm

Portal Vein: 56.1 mm

Confidence: moderate

# L07 · reference liver segment 6

## L07 · automated comparison



Yellow: automated boundary · Red: long axis · Cyan: perpendicular axis · Verify with a radiologist

MARKED DECREASE

LONG AXIS

**15.1 → 9.8 mm**

Change -35.1%

SEGMENTED VOLUME

**1.38 → 0.31 mL**

Change -77.4%

VNC-CORRECTED ENHANCEMENT

**46.4 → 13.1 HU**

Vs local liver: 78.9% → 25.7%

### Automated assessment

Marked automated volume decrease since January.

Minimal enhancement proxy (<10 HU): 0.0% → 33.8%

Absolute <40 HU: 0.0% → 17.6%

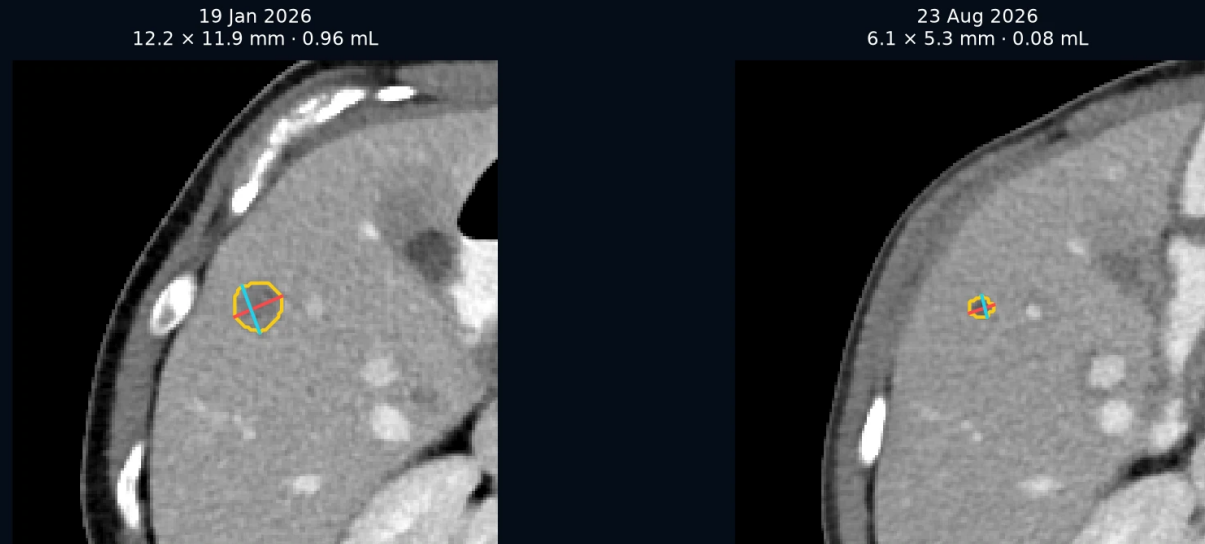
Hepatic Vessels: 2.3 mm

Ivc: 60.2 mm

Confidence: moderate

# L08 · reference liver segment 5

## L08 · automated comparison



Yellow: automated boundary · Red: long axis · Cyan: perpendicular axis · Verify with a radiologist

MARKED DECREASE

LONG AXIS

**12.3 → 6.1 mm**

Change -50.4%

SEGMENTED VOLUME

**0.96 → 0.08 mL**

Change -92.0%

VNC-CORRECTED ENHANCEMENT

**27.9 → 11.2 HU**

Vs local liver: 57.5% → 24.8%

### Automated assessment

Marked automated volume decrease since January.

**Minimal enhancement proxy (<10 HU):** 3.5% → 48.8%

**Absolute <40 HU:** 0.0% → 8.8%

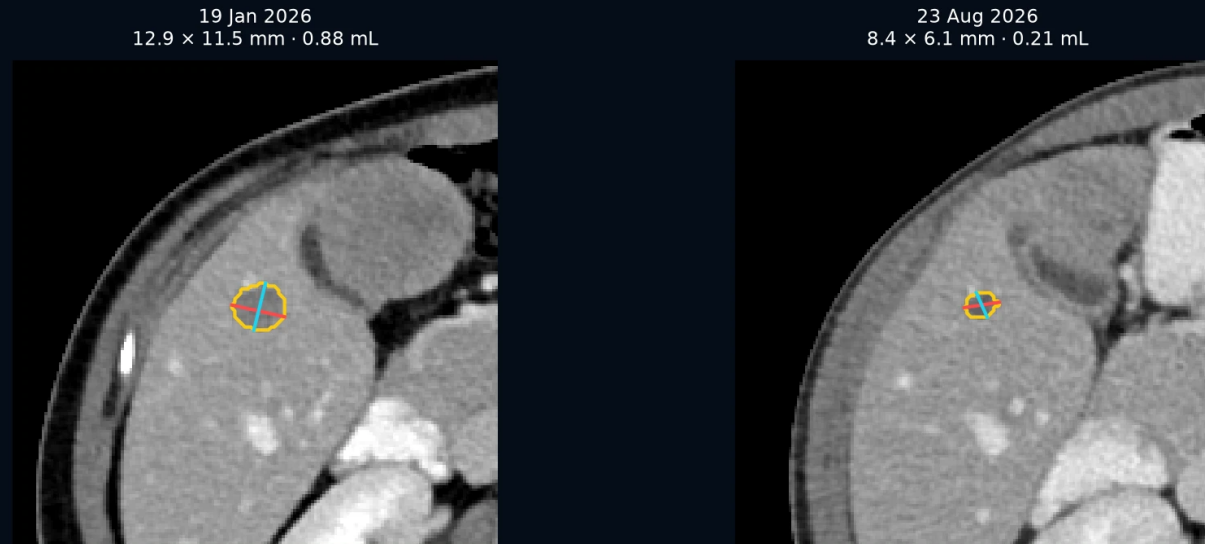
**Hepatic Vessels:** 9.1 mm

**Portal Vein:** 26.1 mm

**Confidence:** low

# L09 · reference liver segment 5

## L09 · automated comparison



Yellow: automated boundary · Red: long axis · Cyan: perpendicular axis · Verify with a radiologist

MARKED DECREASE

LONG AXIS

**12.9 → 8.4 mm**

Change -34.9%

SEGMENTED VOLUME

**0.88 → 0.21 mL**

Change -76.0%

VNC-CORRECTED ENHANCEMENT

**34.4 → -5.6 HU**

Vs local liver: 64.0% → -11.3%

### Automated assessment

Marked automated volume decrease since January.

Minimal enhancement proxy (<10 HU): 1.8% → 93.5%

Absolute <40 HU: 0.0% → 25.8%

Hepatic Vessels: 9.0 mm

Portal Vein: 49.3 mm

Confidence: high

# L10 · reference liver segment 7

## L10 · automated comparison



NOT SEPARATELY DETECTED

LONG AXIS

12.0 → — mm

Change —

SEGMENTED VOLUME

0.71 → 0.00 mL

Change -100.0%

VNC-CORRECTED ENHANCEMENT

27.8 → — HU

Vs local liver: 55.6% → —%

### Automated assessment

Not separately detected by the automated August model; this does not prove complete resolution.

**Confidence:** not separately detected

# L11 · reference liver segment 6

## L11 · automated comparison



NOT SEPARATELY DETECTED

LONG AXIS

**11.6 → — mm**

Change —

SEGMENTED VOLUME

**0.68 → 0.00 mL**

Change -100.0%

VNC-CORRECTED ENHANCEMENT

**32.2 → — HU**

Vs local liver: 62.2% → —%

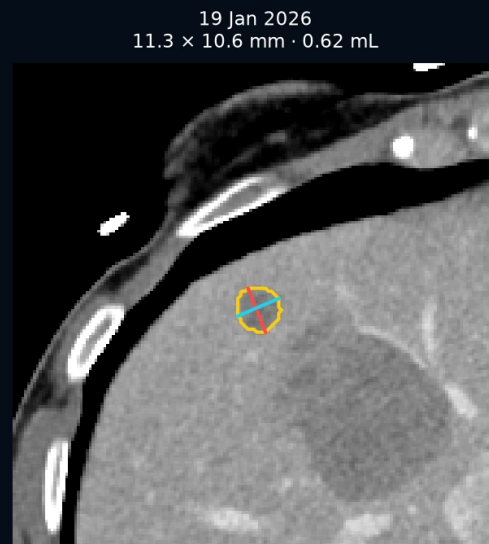
### Automated assessment

Not separately detected by the automated August model; this does not prove complete resolution.

**Confidence:** not separately detected

# L12 · reference liver segment 8

## L12 · automated comparison



Yellow: automated boundary · Red: long axis · Cyan: perpendicular axis · Verify with a radiologist

NOT SEPARATELY DETECTED

LONG AXIS

**11.4 → — mm**

Change —

SEGMENTED VOLUME

**0.62 → 0.00 mL**

Change -100.0%

VNC-CORRECTED ENHANCEMENT

**30.8 → — HU**

Vs local liver: 53.3% → —%

### Automated assessment

Not separately detected by the automated August model; this does not prove complete resolution.

**Confidence:** not separately detected

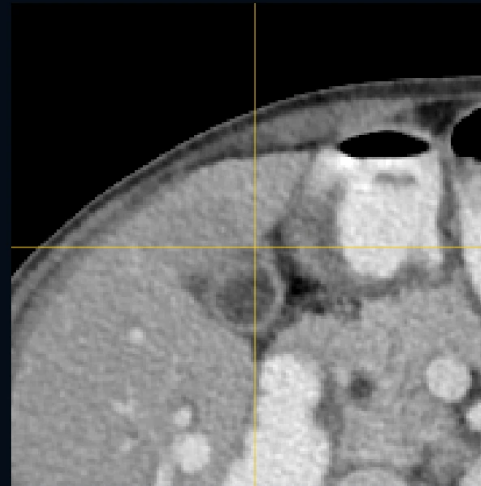
# L13 · reference liver segment 4/5

## L13 · automated comparison

19 Jan 2026  
11.6 × 10.8 mm · 0.48 mL



23 Aug 2026  
No separate automated residual focus



Yellow: automated boundary · Red: long axis · Cyan: perpendicular axis · Verify with a radiologist

NOT SEPARATELY DETECTED

LONG AXIS

**11.8 → — mm**

Change —

SEGMENTED VOLUME

**0.48 → 0.00 mL**

Change -100.0%

VNC-CORRECTED ENHANCEMENT

**30.1 → — HU**

Vs local liver: 62.2% → —%

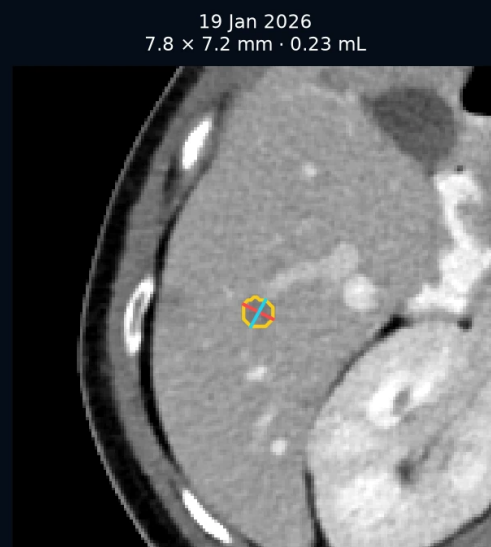
### Automated assessment

Not separately detected by the automated August model; this does not prove complete resolution.

**Confidence:** not separately detected

# L14 · reference liver segment 6

## L14 · automated comparison



Yellow: automated boundary · Red: long axis · Cyan: perpendicular axis · Verify with a radiologist

NOT SEPARATELY DETECTED

LONG AXIS

7.9 → — mm

Change —

SEGMENTED VOLUME

0.23 → 0.00 mL

Change -100.0%

VNC-CORRECTED ENHANCEMENT

32.5 → — HU

Vs local liver: 60.3% → —%

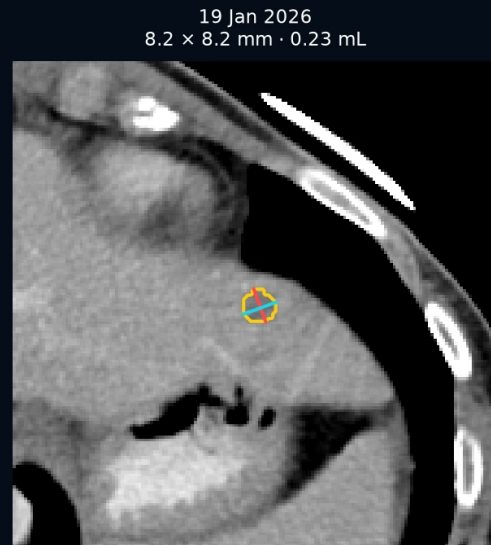
### Automated assessment

Not separately detected by the automated August model; this does not prove complete resolution.

**Confidence:** not separately detected

# L15 · reference liver segment 2

## L15 · automated comparison



Yellow: automated boundary · Red: long axis · Cyan: perpendicular axis · Verify with a radiologist

NOT SEPARATELY DETECTED

LONG AXIS

**8.3 → — mm**

Change —

SEGMENTED VOLUME

**0.23 → 0.00 mL**

Change -100.0%

VNC-CORRECTED ENHANCEMENT

**25.3 → — HU**

Vs local liver: 43.9% → —%

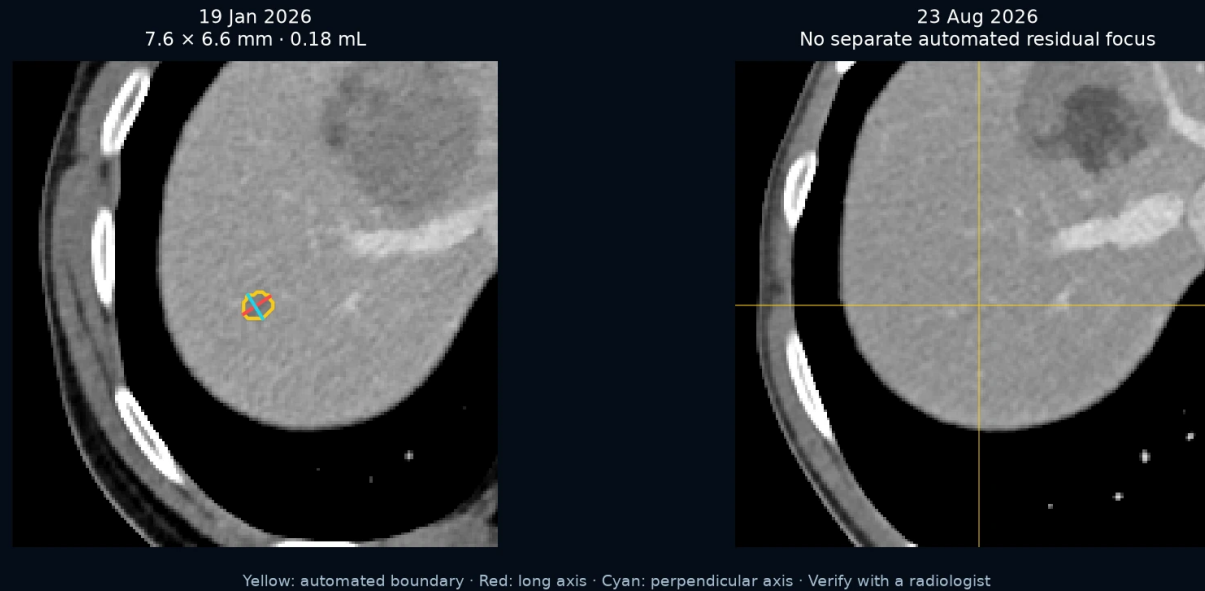
### Automated assessment

Not separately detected by the automated August model; this does not prove complete resolution.

**Confidence:** not separately detected

# L16 · reference liver segment 7

## L16 · automated comparison



NOT SEPARATELY DETECTED

LONG AXIS

**7.7 → — mm**

Change —

SEGMENTED VOLUME

**0.18 → 0.00 mL**

Change -100.0%

VNC-CORRECTED ENHANCEMENT

**34.5 → — HU**

Vs local liver: 67.4% → —%

### Automated assessment

Not separately detected by the automated August model; this does not prove complete resolution.

**Confidence:** not separately detected

Per-lesion January → August changes

| ID  | SEG | JAN MM | AUG MM | JAN ML | AUG ML | VOL Δ   | ENH/LIVER | VESSEL | STATUS       |
|-----|-----|--------|--------|--------|--------|---------|-----------|--------|--------------|
| L01 | 3   | 64.5   | 33.0   | 106.21 | 17.15  | -83.8%  | 12.6      | 2.6    | detected     |
| L02 | 8   | 45.0   | 36.0   | 41.81  | 25.08  | -40.0%  | 63.0      | 0.6    | detected     |
| L03 | 7   | 28.5   | 16.7   | 9.83   | 1.53   | -84.4%  | 23.5      | 14.4   | detected     |
| L04 | 4   | 22.9   | 16.0   | 2.37   | 1.96   | -17.6%  | 87.0      | 1.4    | detected     |
| L05 | 8   | 18.3   | 6.9    | 1.89   | 0.12   | -93.8%  | 60.4      | 4.5    | detected     |
| L06 | 7   | 16.1   | 4.7    | 1.85   | 0.04   | -98.0%  | 72.6      | 13.9   | detected     |
| L07 | 6   | 15.1   | 9.8    | 1.38   | 0.31   | -77.4%  | 25.7      | 2.3    | detected     |
| L08 | 5   | 12.3   | 6.1    | 0.96   | 0.08   | -92.0%  | 24.8      | 9.1    | detected     |
| L09 | 5   | 12.9   | 8.4    | 0.88   | 0.21   | -76.0%  | -11.3     | 9.0    | detected     |
| L10 | 7   | 12.0   | —      | 0.71   | 0.00   | -100.0% | —         | —      | not detected |
| L11 | 6   | 11.6   | —      | 0.68   | 0.00   | -100.0% | —         | —      | not detected |
| L12 | 8   | 11.4   | —      | 0.62   | 0.00   | -100.0% | —         | —      | not detected |
| L13 | 4/5 | 11.8   | —      | 0.48   | 0.00   | -100.0% | —         | —      | not detected |
| L14 | 6   | 7.9    | —      | 0.23   | 0.00   | -100.0% | —         | —      | not detected |
| L15 | 2   | 8.3    | —      | 0.23   | 0.00   | -100.0% | —         | —      | not detected |
| L16 | 7   | 7.7    | —      | 0.18   | 0.00   | -100.0% | —         | —      | not detected |

# How to interpret the automated timeline

## Image-only workflow

Measurements and lesion matching were completed from image pixels and automated masks before any radiologist report review. Original DICOM files and identifiers are not included in this PDF or public app.

## Registration and matching

January was aligned to August using rigid and affine mutual-information registration. Residual foci were linked to prior lesion masks by registered overlap, allowing one older lesion to split into multiple current foci.

## Dimensions and volume

Long and perpendicular axes come from the largest axial contour. Craniocaudal and 3D measurements use voxel spacing. Volume is the sum of segmented voxels and is sensitive to contour boundaries.

## Hemodynamic-aware attenuation

January and August use identical 90-second bolus timing, dose, flow, and 70-keV reconstruction. VNC-corrected lesion enhancement is normalized to local liver and portal-vein enhancement. This reduces—but does not eliminate—circulatory bias. Minimal enhancement and below-40-HU fractions remain appearance proxies, not proof of necrosis or living tumor.

## Vessel proximity

Distances are nearest voxel-to-voxel distances between automated lesion and vessel masks. They are discussion prompts only and are not sufficient for resection, ablation, radiation, or other procedural planning.

## Clinical use

This is not a formal RECIST/mRECIST assessment or medical diagnosis. A radiologist should verify every lesion boundary, segment assignment, match, enhancement measurement, and proximity measurement on the original studies.